1. Show that $\mathbb{P}(A+B+C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) -$

$$- \left[\mathbb{P}(AB) + \mathbb{P}(AC) + \mathbb{P}(BC) \right] +$$

$$+ \mathbb{P}(ABC).$$

Remember: $A+B := A \cup B$, $AB = A \cap B$ in this course.

Could you find a similar identity for $\mathbb{P}(A_1 + A_2 + A_3 + A_4)$?

2. Suppose we have a random permutation of the English alphabet.
What is the probability that at least one of the words bird, fish, rat appear?

3. Two events A and B are called independent if $\mathbb{P}(A|B) = \mathbb{P}(A)$ or $\mathbb{P}(B) = 0$. Show that this is equivalent to saying that

$$\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B).$$

4. In a family of two children there is a girl.
a) What is a probability that both children are girls?

b) Now we know instead that there is a girl who was born in October.
What is the probability that there are two girls in that family?

5. Show that if A is independent of B then A is independent of $\overline{B}$.

5. You have 5 coins in your pocket. Two of them are double-headed, two are normal and one is double tailed (both sides show tail). You take one of them and flip it.

a) What is the probability that it shows head?

b) Suppose it shows head. What is the probability that the other side is head?

c) Now you flip it again. What is the probability that it shows head?

d) Suppose it shows head. What is the probability that the other side is head?